

Economic Fluctuations and the AS–AD Model

Session 4 – Introduction to Macroeconomics

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Recap: Session 3

15 minutes. Answer the questions and complete the exercise silently, on your own.

1. What is the difference between **Nominal** and **Real** GDP?
2. What is **Inflation**, and what are the 3 measures we discussed?
3. What are some **costs of inflation**? Why distinguish **expected** vs **unexpected** inflation? Why might we want a bit of inflation?
4. How should we **compare GDP across countries**? (From video)
5. What is the difference between **economic growth** and **economic development**? (From video)

Exercise: Given Prices, Quantities & CPI Basket

	Base	Year 1	Year 2
Price Bread (P_1)	2	3	4
Price Coffee (P_2)	4	5	5
Quantities ($Q_{1,t}, Q_{2,t}$)	(10, 7)	(12, 9)	(12, 16)
CPI Basket ($C_{1,B}, C_{2,B}$)	(11, 5)	—	—

- Construct Nominal and Real GDP, GDP deflator, CPI, and inflation and weights with both measures

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Inflation is the percentage change in the overall price level.

- ▶ GDP deflator: domestic final production; weights follow output.
- ▶ CPI: a consumer basket, fixed in our example; official weights are updated.
- ▶ PCE deflator: consumption by and on behalf of households.

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Even expected inflation has menu, shoeleather, and tax-related costs. Unexpected inflation also redistributes wealth between borrowers and lenders.

Moderate inflation can ease real-wage adjustment and give monetary policy more room to cut interest rates.

Recap: Answers II

4. How should we compare GDP across countries?

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- ▶ **GDP per capita** adjusts for differences in population size.
- ▶ **Purchasing power parity (PPP)** adjusts for differences in price levels, making purchasing power more comparable.
- ▶ Market exchange rates are useful for financial comparisons, but can mislead about living standards when local prices differ.
- ▶ Even PPP-adjusted GDP per capita is not a complete measure of well-being: distribution, health, and unpaid work also matter.

Recap: Answers III

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- ▶ **Economic growth:** an increase in real GDP; real GDP per capita also accounts for population growth.
- ▶ **Economic development:** broader improvements in well-being, including health, education, institutions, and lower poverty.
- ▶ **HDI** combines income, life expectancy, and education to capture more than production alone.
- ▶ Growth can provide resources for development, but higher GDP does not guarantee that living conditions improve for everyone.

Exercise Data (Two Goods) – Deflator

Compute nominal GDP, real GDP, the GDP deflator, inflation, and the weights.

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Weights (deflator):

$$w_{1,t=1}^{\text{defl}} = \frac{2 \cdot 12}{60} = \boxed{0.40}, \quad w_{2,t=1}^{\text{defl}} = \frac{4 \cdot 9}{60} = \boxed{0.60}; \quad w_{1,t=2}^{\text{defl}} = \boxed{0.27}, \quad w_{2,t=2}^{\text{defl}} = \boxed{0.73}$$

These weights use current quantities valued at base prices. The deflator is a weighted average of price relatives $P_{i,t}/P_{i,B}$; weights change with production.

Exercise Data (Two Goods) – CPI

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Weights (CPI, fixed base basket):

$$w_1^{\text{CPI}} = \frac{P_{1,B} C_{1,B}}{\text{Basket}_B} = \frac{2 \cdot 11}{42} = \boxed{0.524}, \quad w_2^{\text{CPI}} = \frac{P_{2,B} C_{2,B}}{\text{Basket}_B} = \frac{4 \cdot 5}{42} = \boxed{0.476}$$

Prices change, but quantities in this basket are fixed. The CPI is a weighted average of price relatives $P_{i,t}/P_{i,B}$, using constant base-period expenditure shares.

Session 4: Outline

1. **The Short-Run Puzzle**

Fluctuations in output, spending, and employment.

2. **The Long-Run Benchmark**

Classical adjustment and why sticky prices change the outcome.

3. **The AS–AD Framework**

The overall price level and aggregate output.

4. **Aggregate Supply**

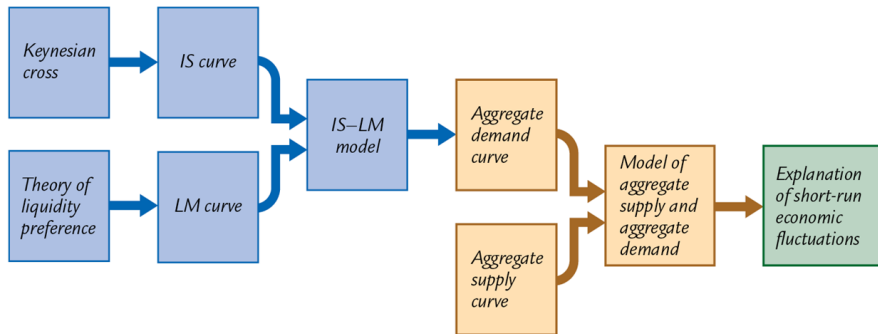
Vertical LRAS and horizontal benchmark SRAS.

5. **From the Short Run to the Long Run**

Demand shocks and the adjustment of prices.

Reading: Mankiw, *Macroeconomics*, 12th ed., Chapter 12; upward-sloping SRAS: Chapter 16.

Our Plan: Sessions 4–8



Mankiw, *Macroeconomics*, 11e, © 2022 Worth Publishers

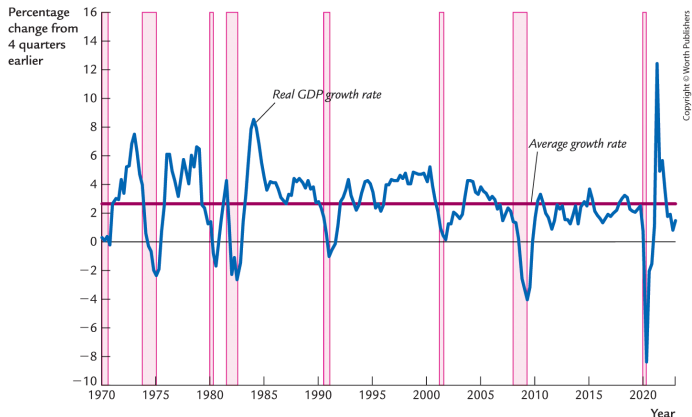
4. **Today:** AS–AD and short-run versus long-run supply.
5. **Keynesian cross and IS:** goods-market equilibrium.
6. **Money:** its role and how its supply is determined.
7. **Liquidity preference and LM:** money demand and interest rates.
8. **Put it together:** IS–LM, derive AD, and apply AS–AD.

Source: Mankiw, *Macroeconomics*, 11th ed. (2022), Worth Publishers.

1. The Short-Run Puzzle

Growth is a long-run trend, not a smooth path

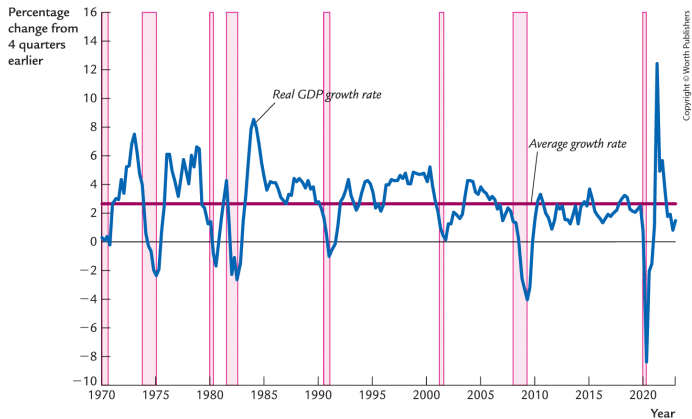
Economic Growth Is Not Smooth



US real GDP grows over the long run, but not smoothly. The horizontal line shows average growth over the period plotted.

Source: Mankiw, *Macroeconomics*, 12th ed. Shading: National Bureau of Economic Research (NBER) recessions.

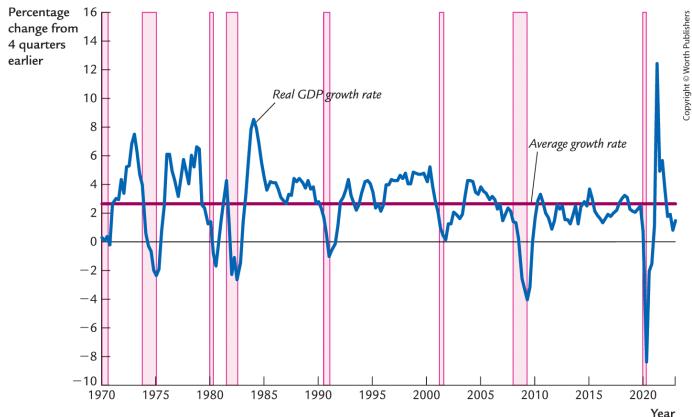
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The average hides large fluctuations: growth sometimes slows sharply or turns negative.

Source: Mankiw, *Macroeconomics*, 12th ed. Shading: National Bureau of Economic Research (NBER) recessions.

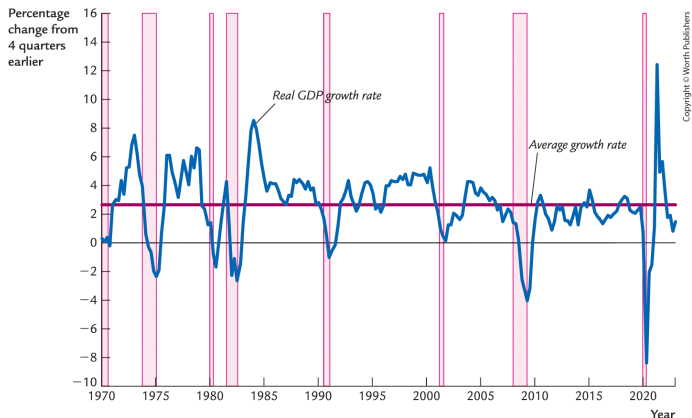
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The shaded periods are NBER recessions. They are **recurring but irregular**: there is no fixed timetable.

Source: Mankiw, *Macroeconomics*, 12th ed. Shading: National Bureau of Economic Research (NBER) recessions.

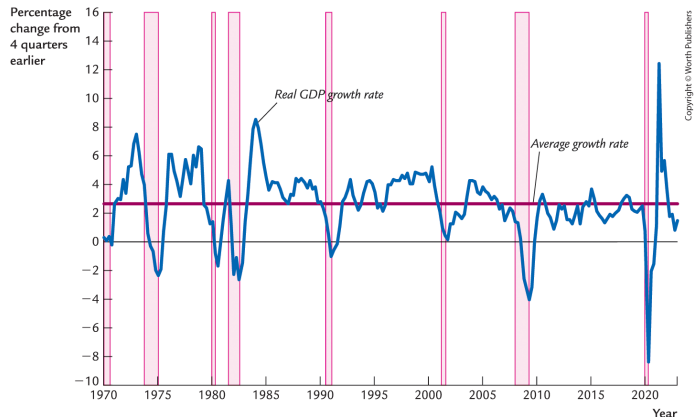
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Peak: the end of an expansion. **Trough:** the end of a recession. The NBER committee dates both using several economy-wide indicators.

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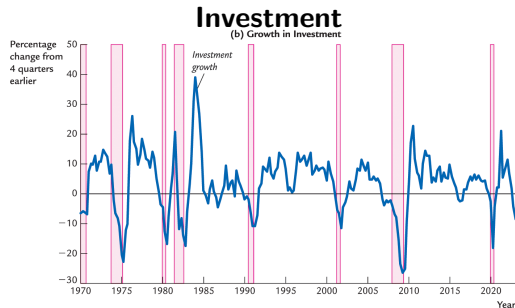
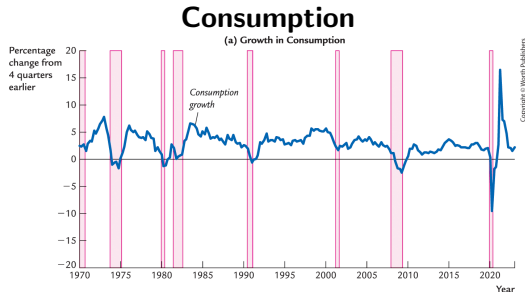
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Our question: Why do recessions happen, and why does output later recover?

Source: Mankiw, *Macroeconomics*, 12th ed. Shading: National Bureau of Economic Research (NBER) recessions.

Consumption and Investment Move With the Cycle



Both weaken in recessions, but **investment is much more volatile**. Notice the different vertical scales.

Source: Mankiw, *Macroeconomics*, 12th ed. Shading: NBER recessions.

The Business Cycle Is Also Visible in Labor Markets



Okun's law: higher unemployment is associated with lower real output relative to its normal path.

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- ▶ Lower demand $\rightarrow$ less output $\rightarrow$ less demand for labor.
- ▶ Lower employment and income $\rightarrow$ less consumption, reinforcing the downturn.

Source: Mankiw, *Macroeconomics*, 12th ed. Shading: NBER recessions.

What Must a Short-Run Model Explain?

1. Why does output fluctuate around a positive long-run growth path?
2. Why do consumption and especially investment weaken in recessions?
3. Why does unemployment rise when output falls?
4. How can policy affect the depth or duration of a downturn?

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Business cycle is the name for these short-run fluctuations. The term does not imply a regular or predictable cycle.

2. The Long-Run Benchmark

What the classical model explains – and what it leaves out

Reminder: How Was Output Determined?

Supply

$$Y^s = F(\bar{K}, \bar{L}) = \bar{Y}$$

- ▶ Capital, labor, and technology determine productive capacity.
- ▶ In the benchmark model, factor supplies and technology are fixed.

Demand

$$Y^d = C(Y - \bar{T}) + I(r) + \bar{G}$$

- ▶ Consumption depends on disposable income $Y - \bar{T}$.
- ▶ The real interest rate r adjusts until desired spending equals output.

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Long-run growth models such as Solow use this supply-side logic while allowing capital, labor, and technology to evolve.

Say's Law and Classical Adjustment

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In a closed economy, r adjusts until planned investment equals national saving: $I(r) = S$.

3. Full employment: flexible wages and competition.

The real wage W/P adjusts to clear the labor market. Excess labor supply puts downward pressure on wages, encouraging hiring.

These are the classical model's adjustment assumptions, not guarantees about every economy. The claim concerns planned spending, not merely the accounting identity $S = I$.

Why Might Price Adjustment Take Time?

The key distinction is price flexibility.

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If demand falls before prices adjust, firms can cut **production and hours worked** instead.

The horizons describe how much adjustment has occurred, not a fixed number of months.

Why Do We Need a Short-Run Theory?

- ▶ The Great Depression exposed the limits of a model that keeps output and employment at their natural levels.
- ▶ In *The General Theory* (1936), **John Maynard Keynes** argued that insufficient demand could sustain low output and high unemployment.
 - ▶ The question is whether interest rates and wages adjust enough to restore full employment.
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"In the long run we are all dead."

John Maynard Keynes

A Tract on Monetary Reform (1923), before the Great Depression. Keynes is criticizing reliance on long-run predictions alone.

With Sticky Prices, Demand Also Matters

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- ▶ **Consumer confidence:** expectations about future income affect consumption.
 - ▶ Fear of losing a job can lead households to postpone purchases and save more today.
- ▶ **Expected profitability:** firms' views about future returns affect investment.
 - ▶ Expecting stronger sales, a firm builds a new factory to increase production.
- ▶ **Monetary and fiscal policy:** financing conditions, government purchases, and taxes affect spending.
 - ▶ Lower interest rates make borrowing cheaper, encouraging investment.
 - ▶ Government purchases raise demand directly; tax cuts can support consumption.

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When spending falls and prices adjust slowly, firms can respond by reducing production and employment.

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This is a rationale for stabilization policy. Its effectiveness depends on the shock, the timing, and how strongly spending responds.

Today we take policy-induced shifts in AD as given. Later sessions explain the mechanisms.

One Policy Change, Two Time Horizons

Suppose the central bank reduces the money supply by **5 percent**.

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Short run: sticky prices

At a given price level, less money raises interest rates and reduces investment: AD shifts left.

Output and employment fall while prices adjust.

Long run: full adjustment

In the simple classical benchmark, the price level and nominal wages fall by 5 percent.

Output, employment, the real wage, and the real interest rate return to their initial levels.

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Monetary neutrality: money does not change long-run real variables.

Classical dichotomy: real and nominal variables can be analyzed separately in that benchmark, but not during sticky-price adjustment.

Holding productive capacity, fiscal policy, and long-run money-demand conditions fixed. Sessions 5–8 explain the interest-rate mechanism.

3. The AS–AD Framework

An economy-wide model of output and the price level

From One Market to the Whole Economy

Microeconomic supply and demand

- ▶ Vertical axis: the price of one good
- ▶ Horizontal axis: quantity of that good
- ▶ Other prices are held constant

Aggregate supply and demand

- ▶ Vertical axis: the economy-wide price level P
- ▶ Horizontal axis: real output Y
- ▶ Goods, labor, and financial markets interact

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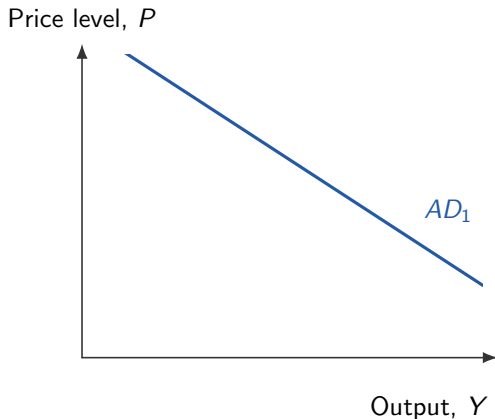
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AS–AD brings supply and demand together to determine the overall price level and real output.

The analogy with a single market is limited: AS–AD combines interactions across **many markets**, rather than simply scaling up supply and demand for one good.

Aggregate Demand: Taken as Given Today

- ▶ The **AD curve** relates the price level to the total quantity of goods and services demanded.
- ▶ It slopes downward: a higher P is associated with less output demanded.
- ▶ A change in P is a **movement along** AD.
- ▶ Changes in spending conditions can **shift** AD.



Intuition: with the money supply fixed, higher prices reduce the purchasing power of money balances, tending to reduce spending. We derive the full mechanism later.

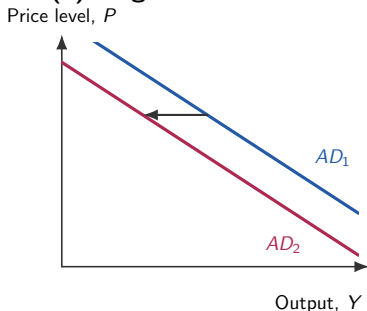
Movement Along AD or Shift of AD?

1. The price level falls, with all other determinants unchanged.
2. Consumer confidence falls, so households reduce spending at every price level.
3. Government purchases increase.

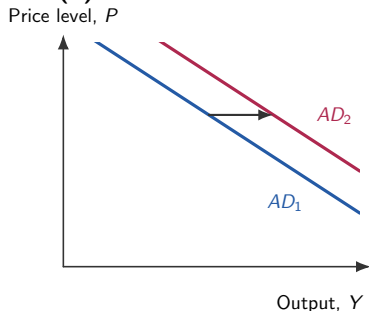
Movement Along AD or Shift of AD?

1. Lower P : a **movement down along** the existing AD curve.
2. Lower consumer confidence: a **leftward shift** of AD.
3. Higher government purchases: a **rightward shift** of AD.

(a) Negative demand shock



(b) Positive demand shock



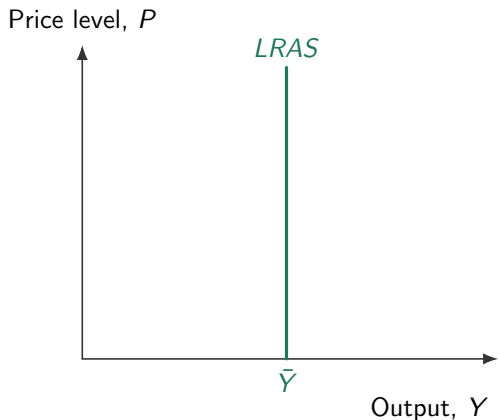
Changes in confidence, fiscal policy, or monetary policy can shift AD. A change in P itself is a movement along the curve.

4. Aggregate Supply

Different price adjustment produces different supply curves

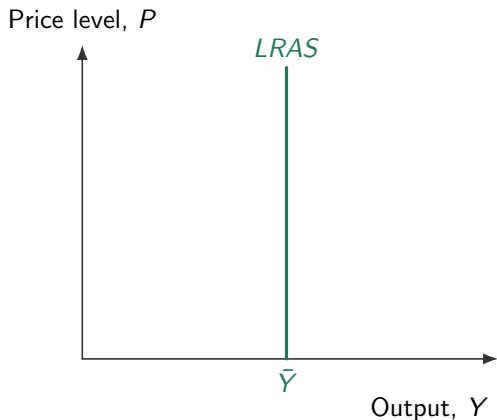
Long-Run Aggregate Supply

- ▶ Plot the benchmark output $\bar{Y}$ at every price level: **LRAS is vertical**.
- ▶ $\bar{Y}$ is **potential** or **full-employment output**: resources are used at sustainable rates.
- ▶ Full employment means unemployment is at its natural rate, not zero.



Long-Run Aggregate Supply

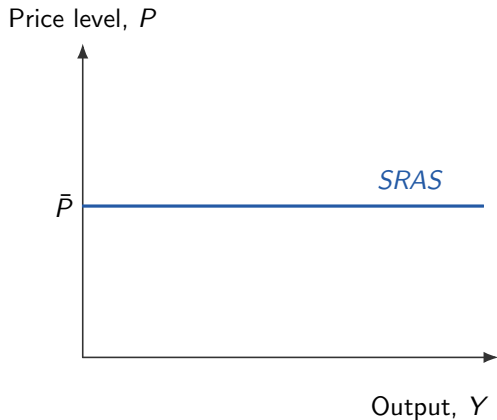
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A shift, not a movement along LRAS: if AI raises productivity, potential output increases and LRAS shifts right.

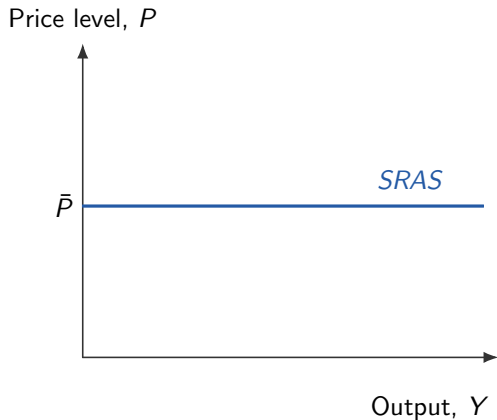
Short-Run Aggregate Supply: The Fixed-Price Benchmark

- ▶ Suppose every firm has issued a **price catalog** that is costly to replace.
- ▶ All prices stay fixed at $P = \bar{P}$: **SRAS is horizontal**.
- ▶ Firms meet demand at those prices by adjusting output and hiring.



Short-Run Aggregate Supply: The Fixed-Price Benchmark

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$\bar{Y}$ is **not a hard ceiling**: overtime and more intensive use of capital can temporarily raise output above its sustainable level.

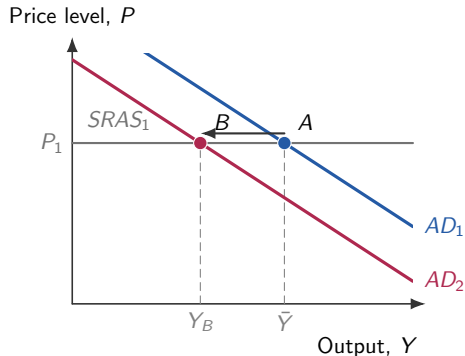
Further reading: Mankiw, 12th ed., Chapter 16.

▶ Upward-sloping SRAS

A Fall in AD: Two Outcomes

Suppose consumer confidence falls. AD shifts left in both diagrams.

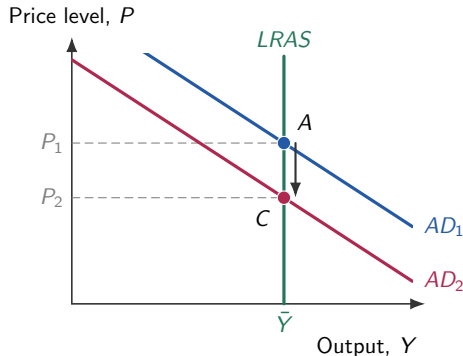
Short run



Output and employment fall; the price level is unchanged.

Both panels start at A and show the same demand shock. B is the short-run outcome; C is the long-run outcome.

Long run



The price level falls; output is back at $\bar{Y}$.

5. From the Short Run to the Long Run

A demand shock and the adjustment of prices

What Is an Economic Shock?

A **shock** is an exogenous change: it originates outside the model and shifts demand or supply. We take the change as given and trace its effects.

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Demand shock: changes spending

- ▶ A collapse in confidence makes households spend less at every price level: **AD shifts left**.
- ▶ A foreign boycott of domestic goods reduces exports: another fall in AD.

Supply shock: changes production

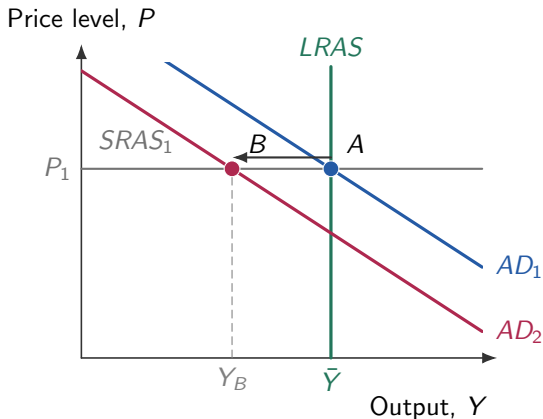
- ▶ An energy-price increase raises production costs: **SRAS shifts up**.
- ▶ A lasting productivity improvement raises potential output: **LRAS shifts right**.

A shock shifts a curve; the resulting changes in output and prices are the economy's response. Not every shock leaves productive capacity unchanged.

A Demand Shock: From the Short Run to the Long Run

Short run: $A \rightarrow B$.

- ▶ Start at A , with output at $\bar{Y}$.
- ▶ A fall in confidence shifts AD left.
- ▶ Prices remain fixed; output falls below $\bar{Y}$.
- ▶ Employment falls; unemployment rises.

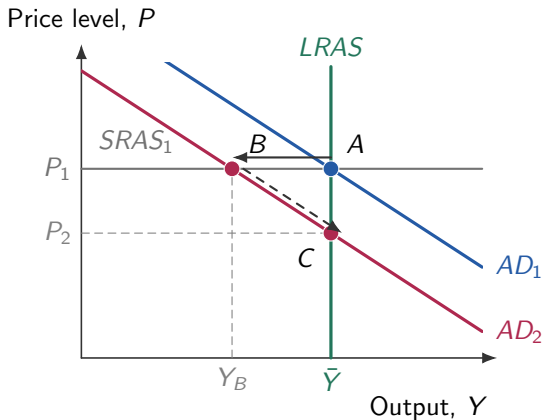


Productive capacity is unchanged. In this combined diagram, B is the short-run outcome and C is the new long-run outcome.

A Demand Shock: From the Short Run to the Long Run

Adjustment: $B \rightarrow C$.

- ▶ Weak demand puts downward pressure on wages and prices.
- ▶ As prices fall, output demanded rises along AD_2 .
- ▶ At C , output returns to $\bar{Y}$, at a lower price level.



Productive capacity is unchanged. In this combined diagram, B is the short-run outcome and C is the new long-run outcome.

One Shock, Two Horizons

	Immediately	After prices adjust
Aggregate demand	Shifts left	Remains at AD_2
Price level	Fixed at P_A	Falls to P_C
Output	Falls below $\bar{Y}$	Returns to $\bar{Y}$
Unemployment	Rises above its natural rate	Returns to its natural rate

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Main lesson: demand determines output while prices are sticky; productive capacity determines output once prices have adjusted.

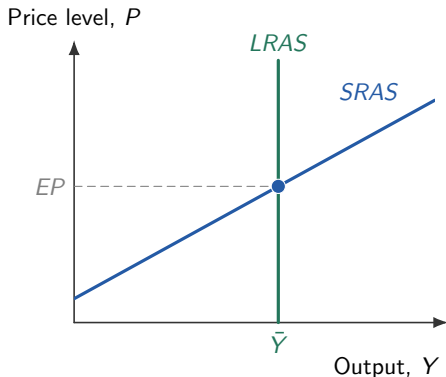
What to Remember

- ▶ **Business cycles are recurring but irregular.** Output, spending, and employment fluctuate together.
- ▶ **Price adjustment distinguishes the two horizons.** LRAS is vertical at potential output; benchmark SRAS is horizontal while prices are fixed.
- ▶ **Analyze the shock and the adjustment separately.** Identify which curve shifts, what is fixed initially, and what changes later.
- ▶ **Demand shocks need not change productive capacity.** Policy may ease the transition; supply shocks can also change potential output.

Next session: the Keynesian cross and the IS curve, our first steps toward constructing aggregate demand.

Appendix: Upward-Sloping SRAS

Allow **some firms to adjust prices**, while expected prices remain fixed.



$$Y = \bar{Y} + \alpha(P - EP), \quad \alpha > 0.$$

EP : expected price level when sticky prices are set.

α : output's response to an unexpected price change.

- ▶ Stronger demand raises output and employment.
- ▶ Expansion raises marginal costs; firms with flexible prices raise their prices.
- ▶ With EP given, output and prices rise together: **SRAS slopes upward.**

With $P > EP$, output exceeds $\bar{Y}$. After expectations and prices fully adjust, $P = EP$ and output returns to $\bar{Y}$.

Further reading: Mankiw, 12th ed., Chapter 16.

[◀ Back to horizontal SRAS](#)